

Knowledge Cutoff Effects on LLM-Generated Literature Reviews in Rapidly Evolving Fields

Abstract

Large Language Models (LLMs) are increasingly deployed to automate scientific literature reviews, promising to synthesize fast-accumulating research across domains such as artificial intelligence, biomedicine, and quantum computing. However, the efficacy of LLM-generated reviews is fundamentally constrained by knowledge cutoffs—the temporal boundary at which pre-training data collection ceases. This paper provides a comprehensive investigation into the mechanisms, manifestations, and mitigations of knowledge cutoff effects on automated academic literature reviews. We demonstrate that knowledge cutoffs do not merely produce static information vacuums regarding post-cutoff publications; they induce complex behavioral distortions, including divergence between reported and effective cutoffs, temporal leakage during ex-ante reasoning, nostalgia and neophilia temporal biases, severe citation hallucination, and evidence-decision decoupling. We evaluate current mitigation paradigms—ranging from standard Retrieval-Augmented Generation (RAG) and paper-centric iterative planning to agentic literature synthesis frameworks and structured graph-based retrieval. Furthermore, we examine algorithmic solutions including Temporal Critique Fine-Tuning (TCFT) and sequentially snapshot-pre-trained language models. Finally, we highlight persistent research challenges in retrieval fragility, context compression, and long-form synthesis evaluation, outlining strategic directions for temporally robust automated scientific discovery.

Keywords

Large Language Models, Automated Literature Reviews, Knowledge Cutoff, Temporal Leakage, Retrieval-Augmented Generation, Scientific Knowledge Synthesis.

Introduction

The exponential growth of scientific literature presents a formidable challenge to academic researchers¹. In rapidly evolving fields such as machine learning, natural language processing, and biomedicine, thousands of preprints and peer-reviewed articles are published monthly across platforms like arXiv, PubMed, and the ACL Anthology¹. To manage this information expansion, researchers have turned to automated literature review systems powered by Large Language Models (LLMs)⁵. These systems aim to automate core stages of the research lifecycle: retrieving relevant literature, extracting methodological contributions, synthesizing thematic taxonomies, and generating structured survey articles⁵.

Despite these advances, LLM-based literature synthesis faces an unavoidable structural constraint: the knowledge cutoff¹. A model's parametric memory reflects only the training corpus available up to its snapshot date¹¹. In rapidly evolving fields where state-of-the-art benchmarks, architectures, and theoretical insights shift over weeks rather than years, relying

on parametric memory introduces severe errors into literature reviews¹. When queried about recent developments, parametric LLMs either completely omit recent breakthroughs or generate plausible but entirely fabricated citations and findings—a phenomenon occurring in 78% to 90% of unaugmented closed-source model outputs on complex literature synthesis tasks³.

Recent attempts to bypass knowledge cutoffs rely primarily on Retrieval-Augmented Generation (RAG) to feed external, up-to-date document passages into the model's context window². However, non-parametric retrieval alone does not resolve the deeper temporal vulnerabilities inherent to LLMs⁸. When models are tasked with historical literature mapping or ex-ante evaluation—reasoning strictly from the standpoint of a specific historical timestamp—they suffer from temporal leakage, exploiting post-cutoff knowledge to answer pre-cutoff queries¹⁰. Furthermore, pre-training corpora suffer from inconsistencies where the "effective cutoff" of specific sub-resources lags significantly behind the model's reported cutoff due to web-crawling temporal noise and deduplication artifacts¹¹.

This research paper presents a systematic analysis of knowledge cutoff dynamics in LLM-generated literature reviews. We analyze the theoretical and empirical failure modes of parametric memory under temporal constraints, evaluate contemporary agentic and structured literature synthesis architectures, and examine emerging algorithmic interventions designed to instill explicit temporal boundary awareness into language models⁸.

Background and Related Work

Automated literature review generation has evolved from basic document summarization into complex, multi-agent orchestrations that emulate human research workflows⁵. The architectures developed to tackle automated literature synthesis can be broadly categorized into four primary paradigms.

One-Shot Static Pipeline Architectures

First-generation survey generation systems, exemplified by early iterations of AutoSurvey, operate on a static, one-shot pipeline paradigm⁵. In these systems, a target research topic serves as an input query⁵. An embedding-based dense retriever fetches a fixed set of document abstracts or full-text passages⁵. An outline-generation LLM analyzes these retrieved documents to construct a global survey outline consisting of sections and subsections⁵. Separate, topic-specific LLM instances then draft section content in parallel based on subsets of the pre-retrieved text, after which a post-processing module merges the sections into a final document⁵.

While computationally efficient, one-shot static pipelines exhibit significant brittleness⁸. Because the initial retrieval step is decoupled from section-level writing, downstream LLM modules receive unstructured, heterogeneous textual context without explicit representations of how research threads, methods, or entities interact across papers². Consequently, the LLM must infer complex domain taxonomies implicitly at generation time, leading to context over-compression, loss of fine-grained evidence, and structural incoherence⁸.

Iterative and Paper-Centric Planning Workflows

To address the rigidity of static outlines, subsequent systems introduced iterative, paper-centric planning mechanisms⁸. Frameworks such as IterSurvey shift from a survey-centric architecture to a paper-centric perspective⁹. Inspired by the human reading process, IterSurvey employs a recurrent outline generation agent that incrementally retrieves, reads, and updates the survey outline as new literature is discovered⁹.

To prevent context dilution, IterSurvey distills each retrieved paper into structured "paper cards" containing explicit summaries of contributions, methodologies, and quantitative findings⁹. Similarly, SurveyForge utilizes a memory-driven Scholar Navigation Agent that performs sub-query decomposition, recursive outline expansion, and temporally-aware reranking over both research-paper databases and prior survey-structure repositories⁸. These iterative approaches improve evidence grounding and reduce structural fragmentation, but remain largely reliant on vector embedding similarity for underlying document discovery⁸.

Agentic Retrieval-Augmented Synthesis

The demand for rigorous, factual scientific synthesis led to the development of autonomous agentic RAG frameworks, most prominently PaperQA2 and OpenScholar⁴. PaperQA2 operationalizes a multi-tool loop that orchestrates paper search, full-text PDF parsing, chunk embedding, citation traversal, evidence scoring, and answer generation¹⁵. By evaluating candidate text passages against specific query requirements using a dedicated Re-scoring and Selection mechanism, PaperQA2 minimizes hallucinations and achieves superhuman performance on scientific question answering, cited topic summarization, and literature contradiction detection¹⁵.

OpenScholar addresses literature synthesis through open-source language models integrated with a specialized paper datastore containing over 45 million open-access papers derived from peS2o⁴. OpenScholar couples dense retrieval with an iterative self-feedback inference loop³. During inference, the system retrieves relevant passages, generates candidate responses, critiques its own output for citation accuracy and factual coverage, and performs target query reformulations to retrieve missing evidence³. On ScholarQABench—a multi-domain benchmark covering computer science, physics, neuroscience, and biomedicine—OpenScholar-8B outperforms proprietary models such as GPT-4o and PaperQA2 in factual correctness and citation attribution³.

Graph-Guided and Structural Retrieval

Recognizing that academic literature reviews inherently organize concepts, methods, tasks, and benchmarks into hierarchical taxonomies, recent architectures introduce explicit structural representations into the retrieval phase². Frameworks like StructSurvey move away from pure text-chunk RAG by dynamically constructing graph-based representations of entities, relations, and taxonomies extracted from retrieved abstracts².

As the survey outline expands, StructSurvey merges extracted subgraphs into an evolving domain knowledge graph². When writing individual sections, downstream LLM agents query this structured graph rather than raw, unstructured text². Empirical evaluations on ACL Anthology benchmark surveys indicate that dynamic graph-guided retrieval significantly improves ROUGE-1 and ROUGE-2 recall, while enhancing LLM-as-a-Judge ratings for logical

structure, depth, and theoretical synthesis².

Main Thematic and Technical Discussion: Anatomy of Knowledge Cutoff Effects

While non-parametric retrieval strategies improve literature grounding, they do not eliminate the operational distortions caused by parametric knowledge cutoffs¹⁰. A rigorous understanding of automated literature review failure requires examining the underlying mechanics of parametric frozen memory, cutoff misalignments, and temporal reasoning failures¹.

Reported versus Effective Cutoffs

Model providers typically publish a single, explicit knowledge cutoff date corresponding to the approximate endpoint of pre-training data collection¹¹. However, empirical probing across data snapshots reveals that the *effective cutoff*—the actual temporal boundary of a model's demonstrated factual recall—varies substantially across sub-resources, domains, and specific topics¹¹.

The divergence between reported and effective cutoffs stems from two primary technical factors during dataset curation¹¹:

1. **Temporal Biases in Web Crawls:** Datasets derived from CommonCrawl snapshots frequently contain legacy web content, stale mirror pages, and re-indexed archival articles¹¹. As a result, a CommonCrawl dump collected in late 2023 may contain substantial proportions of text originating from 2018 or earlier, causing the effective cutoff for niche scientific domains to lag years behind the reported pre-training completion date¹¹.
2. **Deduplication Complications:** Large-scale pre-training pipelines apply semantic and lexical deduplication algorithms to prevent data repetition¹¹. When pre-training corpora combine overlapping snapshots, deduplication schemes frequently remove newer versions of updated documents (or vice versa), leading to arbitrary deletion of temporally sensitive factual updates¹¹.

Consequently, an LLM tasked with writing a literature review may exhibit strong parametric recall for major 2022 machine learning breakthroughs while remaining completely unaware of niche 2021 biomedical discoveries, despite both falling prior to its official cutoff date¹¹.

Mechanisms of Temporal Leakage and Ex-Ante Reasoning Failures

A critical task in academic literature reviews is historical contextualization: tracing how a research field evolved up to a specific historical epoch¹⁰. This task requires *ex-ante reasoning*—the capability to generate responses strictly constrained to information verifiable before a given temporal cutoff timestamp t_c ¹⁰.

Formally, let x denote a literature query and $\mathcal{A}(x)$ represent the universe of factual knowledge associated with x ¹⁰. A temporal mapping function $\tau(f)$ assigns to each fact

$f \in \mathcal{A}(x)$ the exact timestamp when f became publicly verifiable in the scientific record¹⁰.

A temporal cutoff t_c partitions the knowledge space into two disjoint subsets¹⁰:

$$\mathcal{A}_{\text{prev}}(x) = \{f \in \mathcal{A}(x) \mid \tau(f) \leq t_c\}$$

$$\mathcal{A}_{\text{post}}(x) = \{f \in \mathcal{A}(x) \mid \tau(f) > t_c\}$$

The primary failure mode in ex-ante literature review generation is *temporal leakage*: the unintentional inclusion of facts $f \in \mathcal{A}_{\text{post}}(x)$ when the model is explicitly instructed to adhere to cutoff t_c ¹⁰.

Diagnostic analysis of prompt-level interventions reveals fundamental vulnerabilities in how LLMs handle temporal restrictions¹⁰:

- **Phrasing Sensitivity over Semantic Equivalences:** Models evaluate temporal restrictions based on surface-level phrasing rather than logical semantics¹⁰. Explicit knowledge restrictions yield significantly lower temporal leakage rates than logically equivalent implicit historical framings¹⁰.
- **Instruction Positional Bias:** The spatial position of the cutoff instruction within the prompt dominates temporal adherence¹⁰. Placing the cutoff constraint before the query (prefix positioning) reduces temporal leakage far more effectively than placing it after the query (suffix positioning), despite suffix constraints being spatially closer to the generated token sequence¹⁰. Prompt conditioning alters the model's generation trajectory but fails to install an active verification mechanism to check whether candidate facts satisfy $\tau(f) \leq t_c$ ¹⁰.
- **The Temporal Proximity Paradox:** Counterintuitively, temporal leakage is highest for factual events immediately adjacent to the cutoff date¹⁰. In controlled ex-ante evaluations, shorter temporal cutoff gaps produce higher leakage rates, whereas extending the gap significantly reduces leakage¹⁰. This paradox demonstrates that LLMs possess weak fine-grained temporal boundary resolution, struggling most to segregate facts that emerged in the immediate temporal neighborhood of t_c ¹⁰.

Temporal Gap Duration	Question-Answering (QA) Leakage Rate (%)	Factual Adherence Behavior
1-Day Cutoff Gap	40.00%	Severe leakage; high confusion on near-cutoff

		events ¹⁰ .
14-Day Cutoff Gap	32.50%	Moderate leakage; partial separation of temporal boundaries ¹⁰ .
30-Day Cutoff Gap	27.90%	Lowest leakage; clearer demarcation of pre-cutoff knowledge ¹⁰ .

Temporal Degeneration and Generalization Biases

When LLMs are evaluated across dynamic time horizons post-release, they suffer from *temporal degeneration*—a continuous decline in performance, language likelihood, and factual accuracy on newly emerging literature¹. Evaluations on temporal benchmarks such as FreshBench show that high-capacity frontier models degrade rapidly when applied to post-release temporal contexts¹.

This degeneration is exacerbated by parametric temporal biases¹:

1. **Nostalgia Bias:** Prior to the cutoff date, LLMs frequently exhibit a bias toward deeply ingrained, highly repeated historical data, over-indexing on classic paradigms while missing nuanced recent shifts¹.
2. **Neophilia Bias:** Conversely, in the immediate vicinity of the cutoff date, models may over-index on transient, highly publicized pre-cutoff events or preprints that lack long-term scientific validation¹.
3. **Shuffled Corpus Decay:** Standard LLM pre-training shuffles text documents randomly across time¹². Shuffled pre-training creates an implicit temporal delay, causing parametric performance to decay even for facts that strictly preceded the training cutoff¹². Because older facts benefit from greater repetition across web dumps, shuffled models peak on historical data but demonstrate premature decay on near-cutoff knowledge¹².

Evidence-Decision Decoupling in Research Forecasting

Literature reviews are not merely descriptive summaries; they serve as critical foundations for forward-looking research decisions—such as identifying unaddressed scientific bottlenecks, selecting emerging baseline models, or forecasting promising methodological directions²².

Evaluated on benchmark frameworks like ForeSci—which tests whether LLM agents can make forward-looking research judgments using temporally controlled, pre-cutoff knowledge bases—models exhibit a systematic failure termed *evidence-decision decoupling*²².

In evidence-decision decoupling, an LLM agent successfully retrieves and cites relevant pre-cutoff literature evidence, yet fails to synthesize that evidence into the correct forward-looking methodological decision or taxonomic classification²². Explicitly organizing retrieved context into RAG pipelines improves document traceability, but does not resolve the agent's underlying inability to reason about future trajectory directions from incomplete

historical evidence²².

Current Approaches and Developments

To overcome knowledge cutoffs and temporal degradation, the machine learning community has advanced solutions across non-parametric retrieval architectures, structured knowledge graphs, temporal fine-tuning, and sequential pre-training⁸.

Frontier Agentic Literature Synthesis Systems

The state of the art in automated literature review generation relies on specialized agentic RAG frameworks designed to eliminate citation hallucination and maximize evidence coverage⁴.

OpenScholar demonstrates that open-weight models, when paired with specialized retrieval and iterative inference loops, can surpass proprietary models in literature synthesis³. The system operates by executing a structured synthesis workflow: an input scientific query is processed through dense retrieval over a 45-million paper open-access datastore (derived from peS2o)⁴. The retrieved passages are passed into an iterative self-feedback and query refinement loop³. During this loop, the model critiques its candidate draft for missing evidence, reformulates search queries, retrieves additional full-text passages, and updates its text with exact citation attributions, yielding responses with human-level precision³. In blind expert evaluations by PhD researchers across computer science, physics, and biomedicine, OpenScholar-8B responses were preferred over human expert-written literature reviews 51% of the time, while OpenScholar-GPT4o achieved a 70% preference win rate (compared to vanilla GPT-4o's 32%)³.

Similarly, PaperQA2 achieves high factual accuracy on scientific question answering and contradiction detection through structured tool calling¹⁵. PaperQA2 executes full-text search, extracts candidate chunks, re-scores chunk summaries against query requirements via LLM judges, and generates fully cited Wikipedia-style summaries¹⁵. On the LitQA2 benchmark, PaperQA2 matches or exceeds human domain experts in identifying precise scientific findings and detecting contradictory claims across biology literature¹⁹.

Structured Knowledge Graph Retrieval

To prevent context fragmentation inherent to unstructured text chunk retrieval, frameworks like StructSurvey integrate dynamic knowledge graph construction directly into the retrieval pipeline².

The StructSurvey pipeline processes an incoming stream of paper abstracts through an entity and relation extraction module². The extracted entities and relations are integrated into an evolving domain knowledge graph, which interacts directly with a section planning agent². As the outline develops, downstream writing agents execute graph-guided structured generation, ensuring that section drafts are explicitly grounded in extracted methodological and taxonomic relations². This structural grounding yields statistically significant improvements in text generation metrics and human judgment scores for structural coherence, critical analysis, and domain synthesis².

Algorithmic Temporal Grounding Interventions

Beyond non-parametric retrieval, research has advanced algorithmic interventions to instill

explicit temporal boundary awareness directly into model parameters¹⁰.

Temporal Critique Fine-Tuning (TCFT)

Prompt-level constraints modulate temporal conditioning but fail to provide models with temporal verification capabilities¹⁰. To bridge this gap, Temporal Critique Fine-Tuning (TCFT)

trains models to act as explicit temporal verifiers¹⁰. Given a query x , a temporal cutoff t_c , and

a candidate response $\hat{y}$, TCFT trains a student model on teacher-generated temporal critiques¹⁰. The critique identifies post-cutoff leakage, explains temporal boundary violations, and renders an explicit admissibility judgment¹⁰.

Fine-tuning models on temporal critiques installs a functional verification relation between candidate facts and cutoff boundaries¹⁰. When evaluated on ex-ante reasoning tasks, TCFT reduces average temporal leakage by 41.89 percentage points on Qwen2.5-7B-Instruct and 37.79 percentage points on Qwen2.5-14B-Instruct compared to standard supervised fine-tuning (SFT) and prompt baselines¹⁰.

Sequentially Snapshot-Pre-Trained Models (Kairos)

Addressing the temporal opacity caused by shuffled pre-training corpora, the Kairos framework introduces sequentially ordered pre-training¹². Kairos pre-trains language models on chronologically ordered Common Crawl snapshots (e.g., sequentially streaming data from 2020 through 2024)¹².

Comparative evaluations between 6-billion parameter models trained on sequential versus shuffled corpora demonstrate that sequential pre-training preserves general language capabilities while dramatically improving factual freshness and temporal precision¹².

Sequentially pre-trained models exhibit minimal temporal delay decay on recent facts preceding the training cutoff, offering a scalable foundation for continual model updating¹².

Comparative Analysis of Literature Synthesis Architectures

System / Framework	Architectural Paradigm	Primary Knowledge Source	Temporal Grounding Mechanism	Citation Precision / Accuracy	Primary Failure Mode & Limitations
Vanilla LLMs (e.g., GPT-4o, Llama-3) ³	Unaugmented Parametric Generation	Frozen Pre-training Corpus	Static System Prompting / Training Cutoff	Very Low (78%–90% Hallucination Rate) ³	Severe citation hallucination; recency blindness; complete post-cutoff

					ignorance ³ .
AutoSurvey / AutoSurvey2 [cite: 5, 8]	One-Shot Static Pipeline RAG	Vector-indexed Abstracts / Papers	Single-pass Embedding Retrieval at Query Time	Moderate	Context fragmentation; loss of fine-grained evidence; static outline brittleness ⁸ .
IterSurvey [cite: 9]	Paper-Centric Recurrent Planning	Structured Paper Cards & Full Text	Recurrent Outline Updates & Paper-Card Indexing	High	High sequential processing latency; dependency on paper-card distillation quality ⁹ .
PaperQA2 [cite: 15, 18, 19, 20]	Autonomous Multi-Tool Agentic RAG	Full-Text PDFs & Search APIs	Agentic Search, Re-scoring, & Metadata Verification	Superhuman (Matches Expert Precision) ¹⁹	High computational & API cost per query; processing latency on large corpora ¹⁵ .
OpenScholar [cite: 4, 13, 16, 21]	Open-Weight RAG + Iterative Self-Feedback	45M Open-Access Paper Datastore (peS2o)	Self-Feedback Inference Loop & Datastore Indexing	Expert-Level (Exceeds GPT-4o & PaperQA2) ³	Datastore re-indexing overhead; retrieval dependency on open-access availability ⁴ .
StructSurv	Graph-Gui	Extracted	Dynamic	High	Graph

ey [cite: 2, 8]	ded Structural RAG	Entity-Relati on Triples & Abstracts	Graph Updating Aligned with Outline	(Improved ROUGE-1/2 Recall) ²	extraction error propagatio n; domain ontology constructio n latency ² .
TCFT + LLM [cite: 10, 14]	Temporal Critique Fine-Tuned Model	Parametric + Temporal Verification Critic	Explicit Cutoff Admissibilit y Verification	High Temporal Adherence (Leakage Reduced ~40%) ¹⁰	Requires specialized teacher-ge nerated temporal critique datasets for fine-tuning ¹⁰ .
Kairos [cite: 12]	Sequentially Snapshot Pre-Trained Base Model	Chronologi cally Snapshot- Ordered Corpora	Sequential Temporal Snapshot Pre-training	High Factual Freshness	Continuous pre-training infrastructu re requiremen ts; potential catastroph ic forgetting ¹² .

Research Challenges and Limitations

Despite recent architectural progress, key technical challenges hinder the deployment of fully reliable automated literature review systems in rapidly evolving domains⁴.

Retrieval Fragility and Context Compression

Retrieval-Augmented Generation mitigates parametric knowledge cutoffs, but introduces retrieval fragility⁸. Dense vector retrievers frequently struggle to capture complex, multi-hop scientific queries that require synthesizing methodological nuances across disparate sub-fields². When retrieval pipelines return hundreds of full-text paper passages, passing this material into long-context LLMs leads to *context compression* and the "lost-in-the-middle" phenomenon, where models overlook crucial experimental results positioned deep within retrieved contexts⁸.

High Latency and Computational Overhead of Agentic Loops

Frontier agentic synthesis platforms achieve high citation precision by executing complex

iterative loops involving tool calling, paper search, chunk re-scoring, and self-critique³. While effective, these multi-step agent orchestrations impose significant computational overhead and API latency¹⁵. Generating a single comprehensive literature review using agentic RAG can require hundreds of individual LLM calls, making real-time interactive survey generation costly at scale¹⁵.

Evaluation Bottlenecks and Metric Mismatch

Evaluating automated literature reviews remains a major bottleneck². Traditional surface-level lexical overlap metrics (e.g., ROUGE, BLEU) correlate poorly with human expert judgments regarding literature review quality². While pairwise LLM-as-a-Judge protocols (e.g., Survey-Arena) reduce evaluation variance, LLM judges remain prone to length bias, self-preference bias, and difficulty in verifying subtle scientific inaccuracies⁸. Establishing standardized, reproducible benchmarks that combine automated factual verification with human domain expert evaluation remains an ongoing challenge⁴.

Research Gaps and Future Directions

Addressing knowledge cutoff vulnerabilities in automated literature synthesis requires advancing several open research frontiers².

Continual Streaming Pre-Training with Temporal Boundary Preservation

Rather than relying on static pre-training followed by retrieval augmentation, future language models must support continuous, streaming parametric updating¹². Research into sequential snapshot pre-training (e.g., Kairos) must be expanded to develop architecture mechanisms that incorporate new scientific literature in real time without triggering catastrophic forgetting of historical knowledge or eroding fine-grained temporal boundary awareness¹².

Real-Time Dynamic Knowledge Graph Construction

The integration of dynamic knowledge graph construction (e.g., StructSurvey) with high-throughput paper ingestion streams offers a promising pathway for real-time survey generation². Future research should focus on automated, unsupervised ontology learning that continuously extracts, merges, and deduplicates research entities, tasks, and benchmark metrics across preprint repositories, creating living, auto-updating domain knowledge graphs².

Unified Ex-Ante Verification in Agentic Synthesis Workflows

While agentic RAG provides real-time document grounding, integrating explicit temporal critics (e.g., TCFT) into agentic workflows will enable models to execute historical literature synthesis without temporal leakage¹⁰. Coupling non-parametric search tools with parametric temporal critique verifiers will allow automated research systems to generate rigorously verified ex-ante literature reviews across arbitrary historical cutoff dates¹⁰.

Conclusion

Knowledge cutoffs represent a core bottleneck in the application of Large Language Models to automated literature review generation in rapidly evolving fields¹. Parametric knowledge freeze

induces severe failure modes—including reported versus effective cutoff divergence, temporal leakage during historical ex-ante synthesis, temporal degeneration, citation hallucination, and evidence-decision decoupling¹.

Overcoming these limitations requires a shift from static, unaugmented parametric generation toward hybrid architectures that combine agentic search, iterative self-feedback, dynamic graph-guided retrieval, temporal critique fine-tuning, and chronologically ordered sequential pre-training². As these non-parametric grounding mechanisms and temporal verification paradigms mature, automated literature review systems will evolve into accurate, temporally robust partners for scientific discovery.

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